Variables & Equations

ProMo

March 20, 2026

Contents

1 Variables

2 root

	var	symbol	documentation	type	units	eqs
15	$S_{N,q,t}$	S_Nqt	selection matrix or splitter	network		
3	$F^{source}{}_{N,I}$	F_NI_source	incidence matrix NI source	network		
4	$F^{sink}{}_{N,I}$	F_NI_sink	incidence matrix NI sink	network		
2	$F_{N,A}$	F	incidence matrix	network		
11	$I_{t,u}$	I_tu	identity mapping from <t> to <u>	network		
18	cz_I	cz_I	interface variable macro −> control	network		
17	cz_N	cz_N	output from control	network		
14	$S_{N,p,q}$	S_Npu	selection matrix for stacker	network		
6	$F^{sink}{}_{A,I}$	F_AI_sink	incidence matrix AI sink	network		
19	$A_{N,p,q}$	A_Npq	mapping from inputs to outputs	network		
21	$u_{N,t,u}$	u_Ntu	input signal in control domain	network		
13	$S_{I,q}$	S_Aq	selection matrix arcs to outputs	network		

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	var	symbol	documentation	type	units	eqs
16	mv_I	`mv_I`	interface variable macro −> control	network		
10	$S_{I,q}$	`S_Iq`	selection matrix interface to control output	network		
20	$A_{N,t,u}$	`A_Ntu`	mapping from input elements to outputs	network		
22	$y_{N,t,u}$	`y_Ntu`	output signal in control domain	network		
9	$S_{I,p}$	`S_Ip`	selection matrix interface to control input	network		
7	${F^{source}}_{N,A}$	`F_NA_source`	incidence matrix NA source	network		
12	$S_{A,p}$	`S_Ap`	selection matrix interface species-related measures	network		
8	${F^{sink}}_{N,A}$	`F_NA_sink`	incidence matrix NA sink	network		
27	$I_{N,A}$	`I_NA`	identity mapping from <N> to <A>	network		
5	${F^{source}}_{A,I}$	`F_AI_source`	incidence matrix AI source	network		
105	t^o	`to`	starting time	frame	s	4
1	t	`t`	time	frame	s	
107	Δt	`t_interval`	time interval	frame	s	6
106	t^e	`te`	end time	frame	s	5

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	var	symbol	documentation	type	units	eqs
103	1	`one`	numerical value one	constant		2
102	0	`zero`	numerical value zero	constant		1
104	0.5	`oneHalf`	numerical value one half	constant		3
101	#	`value`	numerical value	constant		

3 physical

	var	symbol	documentation	type	units	eqs
25	$r_{z\,N}$	`r_z`	z-coordinate	frame	m	
24	$r_{y\,N}$	`r_y`	y-coordinate	frame	m	
23	$r_{x\,N}$	`r_x`	x-coordinate	frame	m	
108	U_N	`U`	fundamental state – internal energy	state	$kg\,m^2\,s^{-2}$	
111	$n_{N,S}$	`n`	fundamental state – molar mass	state	mol	93
137	m_N	`m`	mass	state	kg	30
109	S_N	`S`	fundamental state – internal entropy	state	$kg\,m^2\,K^{-1}\,s^{-2}$	
144	C_N	`C`	fundamental state – charge	state	$A\,s$	
110	V_N	`V`	volume	state	m^3	7
123	R	`R`	gas constant	constant	$kg\,m^2\,mol^{-1}\,K^{-1}\,s^{-2}$	17
121	N^A	`Avo`	Avogadro constant	constant	mol^{-1}	
132	λ_S	`Mm`	molecular masses	constant	$kg\,mol^{-1}$	
122	k^B	`Boltz`	Boltzmann constant	constant	$kg\,m^2\,K^{-1}\,s^{-2}$	

Continued on next page

	var	symbol	documentation	type	units	eqs
150	$A_{yz\,N}$	`Ayz`	cross sectional area yz	secondaryState	m^2	42
149	$A_{xz\,N}$	`Axz`	cross sectional are xz	secondaryState	m^2	41
143	ρ_N	`rho`	density	secondaryState	$kg\,m^{-3}$	36
148	$A_{xy\,N}$	`Axy`	cross sectional area xy	secondaryState	m^2	40

4 macroscopic

	var	symbol	documentation	type	units	eqs
206	$\dot{H}^{d}_{y\,N}$	aHnd_y	accumulation of enthalpy due to diffusional mass flow in y-direction	transport	$kg\,m^{2}\,s^{-3}$	101
151	$\hat{q}_{x\,A}$	fq_x	heat flow in x-direction	transport	$kg\,m^{2}\,s^{-3}$	43
194	$\dot{n}^{c}_{x\,N,S}$	anc_x	accumulation of molar mass due to convection	transport	$mol\,s^{-1}$	87
155	$\hat{n}^{d}_{y\,A,S}$	fnd_y	diffusion flow in y-direction	transport	$mol\,s^{-1}$	47 90
214	$\dot{w}_{N}$	aw	accumulation of enthalpy due to work flow	transport	$kg\,m^{2}\,s^{-3}$	109
156	$\hat{n}^{d}_{z\,A,S}$	fnd_z	diffusion flow in z-direction	transport	$mol\,s^{-1}$	48 91
158	$c_{A,S}$	c_AS	concentration in convective event-dynamic flow	transport	$m^{-3}\,mol$	50
157	d_{A}	d	flow direction of convective flow	transport		49
152	$\hat{q}_{y\,A}$	fq_y	heat flow in y-direction	transport	$kg\,m^{2}\,s^{-3}$	44
212	$\dot{n}^{d}_{y\,N,S}$	and_y	accumulation due to diffusion in y-direction	transport	$mol\,s^{-1}$	107
160	$\hat{n}^{c}_{x\,A,S}$	fnc_x	molar convective flow in x-direction	transport	$mol\,s^{-1}$	52
154	$\hat{n}^{d}_{x\,A,S}$	fnd_x	diffusion flow in x-direction	transport	$mol\,s^{-1}$	46 89
234	$\hat{m}_{N,A}$	fm	convective mass flow	transport	$kg\,s^{-1}$	133

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	var	symbol	documentation	type	units	eqs
195	$\dot{n}^{d}_{x}{}_{N,S}$	`and_x`	accumulation due to diffusion in x-direction	transport	$mol\, s^{-1}$	88
153	$\hat{q}_{z}{}_{A}$	`fq_z`	heat flow in z-direction	transport	$kg\, m^{2}\, s^{-3}$	45
210	$\dot{q}_{z}{}_{N}$	`aq_z`	accumulation due to heat flow in z-direction	transport	$kg\, m^{2}\, s^{-3}$	105
211	$\hat{w}_{A}$	`fw`	a fixed work flow to start with	transport	$kg\, m^{2}\, s^{-3}$	106
209	$\dot{q}_{y}{}_{N}$	`aq_y`	accumulation due to heat flow in y-direction	transport	$kg\, m^{2}\, s^{-3}$	104
205	$\dot{H}^{d}_{x}{}_{N}$	`aHnd_x`	accumulation of enthalpy due to diffusional mass flow in x-direction	transport	$kg\, m^{2}\, s^{-3}$	100
208	$\dot{q}_{x}{}_{N}$	`aq_x`	accumulation due to heat flow in x-direction	transport	$kg\, m^{2}\, s^{-3}$	103
159	$\hat{V}_{A}$	`fV`	volumetric flow in x-direction	transport	$m^{3}\, s^{-1}$	51
213	$\dot{n}^{d}_{z}{}_{N,S}$	`and_z`	accumulation due to diffusion in z-direction	transport	$mol\, s^{-1}$	108
207	$\dot{H}^{d}_{z}{}_{N}$	`aHnd_z`	accumulation of enthalpy due to diffusional mass flow in z-direction	transport	$kg\, m^{2}\, s^{-3}$	102
204	$\dot{H}^{c}_{x}{}_{N}$	`aHnc_x`	accumulation of enthalpy due to convective mass flow in x-direction	transport	$kg\, m^{2}\, s^{-3}$	99
191	$\hat{k}^{d,Fick}_{y}{}_{A,S}$	`kdAFick_y`	Fick diffusivity in arc and y-direction	properties	ms^{-1}	84
219	$R^{e}{}_{N}$	`elResistant`	electrical resistant	properties	$kg\, m^{2}\, A^{-2} s^{-3}$	115
189	ρ_{A}	`rhoA`	density in arc	properties	$kg\, m^{-3}$	82

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	var	symbol	documentation	type	units	eqs
186	${k^q_x}_A$	kqA_x	thermal conductivity in arc and x-direction	properties	$kg\,K^{-1}\,s^{-3}$	79
185	${k^c_z}_A$	kcA_z	convecive mass conductivity in arc and y-direction	properties	$m^{-1}\,s$	78
183	${k^c_x}_A$	kcA_x	convective mass conductivity in arc and x diretion	properties	$m^{-1}\,s$	76
187	${k^q_y}_A$	kqA_y	thermal conductivity in arc and y-direction	properties	$kg\,K^{-1}\,s^{-3}$	80
188	${k^q_z}_A$	kqA_z	thermal conductivity in arc and z-direction	properties	$kg\,K^{-1}\,s^{-3}$	81
193	$h_{A,S}$	hA	partial molar enthalpiies in arc	properties	$kg\,m^2\,mol^{-1}\,s^{-2}$	86
192	${\hat{k}^{d,Fick}_z}_{A,S}$	kdAFick_z	Fick diffusivity in arc and z-direction	properties	ms^{-1}	85
181	${k^d_y}_{A,S}$	kdA_y	diffusivity in arc and y-direction	properties	$kg^{-1}\,m^{-4}\,mol^2\,s$	74
182	${k^d_z}_{A,S}$	kdA_z	diffusivity in arc and z-direction	properties	$kg^{-1}\,m^{-4}\,mol^2\,s$	75
184	${k^c_y}_A$	kcA_y	convective mass conductivity in arc and y-direction	properties	$m^{-1}\,s$	77
190	${\hat{k}^{d,Fick}_x}_{A,S}$	kdAFick_x	Fick's diffusivity in arc and x-direction	properties	ms^{-1}	83
180	${k^d_x}_{A,S}$	kdA_x	diffusivity in arc and x-direction	properties	$kg^{-1}\,m^{-4}\,mol^2\,s$	73
117	G_N	G	Gibbs free energy	state	$kg\,m^2\,s^{-2}$	13
116	A_N	A	Helmholtz energy	state	$kg\,m^2\,s^{-2}$	12

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	var	symbol	documentation	type	units	eqs
115	H_N	`H`	Enthalpy	state	$kg\, m^2\, s^{-2}$	11 112
216	$H^o{}_N$	`Ho`	initial enthalpy	state	$kg\, m^2\, s^{-2}$	111
203	$n^o{}_{N,S}$	`no`	initial mass	state	mol	98
114	$\mu_{N,S}$	`chemPot`	chemical potential	effort	$kg\, m^2\, mol^{-1}\, s^{-2}$	10 54
161	$\mu^o{}_{N,S}$	`chemPotStandard`	instantiating standard chemical potential	effort	$kg\, m^2\, mol^{-1}\, s^{-2}$	53
113	T_N	`T`	temperature	effort	K	9 121
217	$U^e{}_N$	`Ue`	electrical potential – voltage	effort	$kg\, m^2\, A^{-1} s^{-3}$	113
112	p_N	`p`	thermodynamic pressure	effort	$kg\, m^{-1}\, s^{-2}$	8
125	$C_{V\,N}$	`CV`	total heat capacity at constant volume	secondaryState	$kg\, m^2\, K^{-1}\, s^{-2}$	19
138	$c_{N,S}$	`c`	molar concentration	secondaryState	$m^{-3}\, mol$	31
119	$v_{y\,N}$	`v_y`	velocity in y-direction	secondaryState	ms^{-1}	15
141	$c_{p\,N}$	`cp`	specific heat capacity at constant pressure	secondaryState	$m^2\, K^{-1}\, s^{-2}$	34 120
136	$h_{N,S}$	`h`	partial molar enthalpies	secondaryState	$kg\, m^2\, mol^{-1}\, s^{-2}$	29
142	$c_{V\,N}$	`cV`	specific heat capacity at constant volume	secondaryState	$m^2\, K^{-1}\, s^{-2}$	35

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	var	symbol	documentation	type	units	eqs
139	${n^t}_N$	nt	total number of moles	secondaryState	mol	32
124	${C_p}_N$	Cp	total heat capacity at constant pressure	secondaryState	$kg\,m^2\,K^{-1}\,s^{-2}$	18 117
222	${T^{ref}}_N$	T_ref	reference temperature	secondaryState	K	119
140	$x_{N,S}$	x	mole fraction	secondaryState		33
120	${v_z}_N$	v_z	velocity in z-direction	secondaryState	ms^{-1}	16
118	${v_x}_N$	v_x	velocity in x-direction	secondaryState	ms^{-1}	14
202	$\tilde{n}_{N,S}$	np	link variable np to interface macroscopic	conversion	$m^{-3}\,mol\,s^{-1}$	97
196	$\dot{n}_{N,S}$	an	differential mass balance without reaction	diffState	$mol\,s^{-1}$	92
215	$\dot{H}_N$	dH	accumulation of enthalpy	diffState	$kg\,m^2\,s^{-3}$	110
220	${\dot{U}^e}_A$	dUe	Kirkhoffs first law	diffState	$kg\,m^2\,A^{-1}s^{-3}$	116
218	${I^e}_N$	current	current definition	internalTransport	A	114
224	$\bar{T}_N$	T_meas	temperature measurement	observation		123
223	${T^n}_N$	T_meas_norming	value to norm measurement of temperature	observation	K	122

5 reactions

	var	symbol	documentation	type	units	eqs
198	$K^{o}{}_{K}$	`Ko`	Arrhenius frequency factor	constant	$m^{-3}\,mol\,s^{-1}$	
26	$N_{S,K}$	`N`	stoichiometric matrix	constant		
197	$E^{a}{}_{K}$	`Ea`	Arrhenius activation energy	constant	$kg\,m^{2}\,mol^{-1}\,s^{-2}$	
167	$T_{N,p}$	`T`	link variable T to interface reactions	effort	K	60
165	$x_{N,S,p}$	`x`	link variable x to interface reactions	secondaryState		58
163	$c_{N,S,p}$	`c`	link variable c to interface reactions	secondaryState	$m^{-3}\,mol$	56
169	$\xi_{N,K,p}$	`probability`	probability of reaction to take place	conversion		62
199	$K_{N,K,p}$	`K`	Arrhenius reaction "constant"	conversion	$m^{-3}\,mol\,s^{-1}$	94
200	$\tilde{n}_{N,S,q}$	`np`	production from reaction set	conversion	$m^{-3}\,mol\,s^{-1}$	95
168	$f_{N,S,K,p}$	`factor`	factor for probability computation	conversion		61

6 control

	var	symbol	documentation	type	units	eqs
245	$T_meas_{N,p}$	`T_meas`	link variable mv I to interface control	algebraic		144

7 macroscopic–reactions

	var	symbol	documentation	type	units	eqs
162	$\_c_{I,S}$	`_c`	link variable c to interface macroscopic »> reactions with source:node	get	$m^{-3}\,mol$	55
164	$\_x_{I,S}$	`_x`	link variable x to interface macroscopic »> reactions with source:node	get		57
166	$\_T_{I}$	`_T`	link variable T to interface macroscopic »> reactions with source:node	get	K	59

8 macroscopic–control

	var	symbol	documentation	type	units	eqs
244	$\_mv\_I_{I}$	`_mv_I`	link variable mv I to interface macroscopic »> control with source:node	get		143

9 reactions–macroscopic

	var	symbol	documentation	type	units	eqs
201	$\_np_{I,S}$	`_np`	link variable np to interface reactions »> macroscopic with source:node	get	$m^{-3}\,mol\,s^{-1}$	96

10 Equations

11 Generic

no	equation	documentation	layer
1	$0 := \textbf{Instantiate}(\#, \#)$	numerical value zero	root
2	$1 := \textbf{Instantiate}(\#, \#)$	numerical value one	root
3	$0.5 := \textbf{Instantiate}(\#, \#)$	numerical value one half	root
4	$t^o := \textbf{Instantiate}(t, \#)$	starting time	root
5	$t^e := \textbf{Instantiate}(t, \#)$	end time	root
6	$\Delta t := \textbf{Instantiate}(t, \#)$	time interval	root
7	$V_N := r_{xN} \,.\, r_{yN} \,.\, r_{zN}$	volume	physical
8	$p_N := \frac{\partial U_N}{\partial V_N}$	thermodynamic pressure	physical
9	$T_N := \frac{\partial U_N}{\partial S_N}$	temperature	macroscopic
10	$\mu_{N,S} := \frac{\partial U_N}{\partial n_{N,S}}$	chemical potential	macroscopic
11	$H_N := U_N - p_N \,.\, V_N$	Enthalpy	macroscopic
12	$A_N := U_N - T_N \,.\, S_N$	Helmholtz energy	macroscopic
13	$G_N := U_N + p_N \,.\, V_N - T_N \,.\, S_N$	Gibbs free energy	macroscopic
14	$v_{xN} := \frac{\partial r_{xN}}{\partial t}$	velocity in x-direction	macroscopic
15	$v_{yN} := \frac{\partial r_{yN}}{\partial t}$	velocity in y-direction	macroscopic

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no	equation	documentation	layer
16	$v_{zN} := \frac{\partial r_{zN}}{\partial t}$	velocity in z-direction	macroscopic
17	$R := N^{A} \, . \, k^{B}$	gas constant	physical
18	$C_{pN} := \frac{\partial H_N}{\partial T_N}$	total heat capacity at constant pressure	macroscopic
19	$C_{VN} := \frac{\partial U_N}{\partial T_N}$	total heat capacity at constant volume	macroscopic
29	$h_{N,S} := H_N \, . \, (n_{N,S})^{-1}$	partial molar enthalpies	macroscopic
30	$m_N := \lambda_S \overset{S}{\star} n_{N,S}$	mass	macroscopic
31	$c_{N,S} := (V_N)^{-1} \, . \, n_{N,S}$	molar concentration	macroscopic
32	$n^t{}_N := \textbf{reduceSum}\,(n_{N,S}, S)$	total number of moles	macroscopic
33	$x_{N,S} := (n^t{}_N)^{-1} \, . \, n_{N,S}$	mole fraction	macroscopic
34	$c_{pN} := C_{pN} \, . \, (m_N)^{-1}$	specific heat capacity at constant pressure	physical
35	$c_{VN} := C_{VN} \, . \, (m_N)^{-1}$	specific heat capacity at constant volume	macroscopic
36	$\rho_N := (V_N)^{-1} \, . \, m_N$	density	physical
40	$A_{xyN} := r_{xN} \, . \, r_{yN}$	cross sectional area xy	physical
41	$A_{xzN} := r_{xN} \, . \, r_{zN}$	cross sectional are xz	physical
42	$A_{yzN} := r_{yN} \, . \, r_{zN}$	cross sectional area yz	physical
43	$\hat{q}_{xA} := k^q_{xA} \, . \, A_{yzN} \, . \, F_{N,A} \overset{N}{\star} T_N$	heat flow in x-direction	macroscopic
44	$\hat{q}_{yA} := k^q_{yA} \, . \, A_{xzN} \, . \, F_{N,A} \overset{N}{\star} T_N$	heat flow in y-direction	macroscopic

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no	equation	documentation	layer
45	$\hat{q}_{z\,A} := k^{q}_{z\,A} \,.\, A_{xy\,N} \,.\, F_{N,A} \overset{N}{\star} T_{N}$	heat flow in z-direction	macroscopic
46	$\hat{n}^{d}_{x\,A,S} := \hat{k}^{d,Fick}_{x}{}_{A,S} \,.\, A_{yz\,N} \,.\, F_{N,A} \overset{N}{\star} c_{N,S}$	Fick diffusion flow in x-direction	macroscopic
47	$\hat{n}^{d}_{y\,A,S} := \hat{k}^{d,Fick}_{y}{}_{A,S} \,.\, A_{xz\,N} \,.\, F_{N,A} \overset{N}{\star} c_{N,S}$	Fick diffusion flow in y-direction	macroscopic
48	$\hat{n}^{d}_{z\,A,S} := \hat{k}^{d,Fick}_{z}{}_{A,S} \,.\, \left(A_{xy\,N} \,.\, F_{N,A}\right) \overset{N}{\star} c_{N,S}$	Fick diffusion flow in z-direction	macroscopic
49	$d_{A} := \mathbf{sign}\left(F_{N,A} \overset{N}{\star} p_{N}\right)$	flow direction of convective flow	macroscopic
50	$c_{A,S} := \left(0.5 \,.\, \left(F_{N,A} - d_{A} \,.\, \lvert F_{N,A}\rvert\right)\right) \overset{N}{\star} c_{N,S}$	concentration in convective event-dynamic flow	macroscopic
51	$\hat{V}_{A} := \left(\rho_{A}\right)^{-1} \,.\, k^{c}_{x\,A} \,.\, A_{yz\,N} \,.\, F_{N,A} \overset{N}{\star} p_{N}$	volumetric flow in x-direction	macroscopic
52	$\hat{n}^{c}_{x\,A,S} := \hat{V}_{A} \,.\, c_{A,S}$	molar convective flow in x-direction	macroscopic
53	$\mu^{o}{}_{N,S} := \mathbf{Instantiate}(\mu_{N,S}, \#)$	instantiating standard chemical potential	macroscopic
54	$\mu_{N,S} := \mu^{o}{}_{N,S} + R \,.\, T_{N} \,.\, \mathbf{ln}\left(x_{N,S}\right)$	chemical potential standard model with mole fraction	macroscopic
61	$f_{N,S,K,p} := x_{N,S,p}{}^{((\lvert N_{S,K}\rvert))}$	factor for probability computation	reactions
62	$\xi_{N,K,p} := \prod_{S} f_{N,S,K,p}$	probability of reaction to take place	reactions
73	$k^{d}_{x\,A,S} := I_{N,A} \overset{N}{\star} \left(\left(\mu_{N,S}\right)^{-1} \,.\, \left(v_{x\,N} \,.\, \left(\left(V_{N}\right)^{-1} \,.\, \frac{\partial U_{N}}{\partial \mu_{N,S}}\right)\right)\right)$	diffusivity in arc and x-direction	macroscopic
74	$k^{d}_{y\,A,S} := I_{N,A} \overset{N}{\star} \left(\left(\mu_{N,S}\right)^{-1} \,.\, \left(v_{y\,N} \,.\, \left(\left(V_{N}\right)^{-1} \,.\, \frac{\partial U_{N}}{\partial \mu_{N,S}}\right)\right)\right)$	diffusivity in arc and y-direction	macroscopic
75	$k^{d}_{z\,A,S} := I_{N,A} \overset{N}{\star} \left(\left(\mu_{N,S}\right)^{-1} \,.\, \left(v_{z\,N} \,.\, \left(\left(V_{N}\right)^{-1} \,.\, \frac{\partial U_{N}}{\partial \mu_{N,S}}\right)\right)\right)$	diffusivity in arc and z-direction	macroscopic
76	$k^{c}_{x\,A} := I_{N,A} \overset{N}{\star} \left(\left(\lambda_{S} \overset{S}{\star} \left(\mu_{N,S}\right)^{-1}\right) \,.\, \left(V_{N}\right)^{-1} \,.\, \frac{\partial U_{N}}{\partial p_{N}} \,.\, v_{x\,N}\right)$	convective mass conductivity in arc and x diretion	macroscopic

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no	equation	documentation	layer
77	$k^c_{y\,A} := I_{N,A} \overset{N}{\star} \left(\left(\lambda_S \overset{S}{\star} (\mu_{N,S})^{-1}\right) . (V_N)^{-1} . \frac{\partial U_N}{\partial p_N} . v_{y\,N}\right)$	convective mass conductivity in arc and y-direction	macroscopic
78	$k^c_{z\,A} := I_{N,A} \overset{N}{\star} \left(\left(\lambda_S \overset{S}{\star} (\mu_{N,S})^{-1}\right) . (V_N)^{-1} . \frac{\partial U_N}{\partial p_N} . v_{z\,N}\right)$	convecive mass conductivity in arc and y-direction	macroscopic
79	$k^q_{x\,A} := I_{N,A} \overset{N}{\star} \left((V_N)^{-1} . C_{p\,N} . v_{x\,N}\right)$	thermal conductivity in arc and x-direction	macroscopic
80	$k^q_{y\,A} := I_{N,A} \overset{N}{\star} \left((V_N)^{-1} . C_{p\,N} . v_{y\,N}\right)$	thermal conductivity in arc and y-direction	macroscopic
81	$k^q_{z\,A} := I_{N,A} \overset{N}{\star} \left((V_N)^{-1} . C_{p\,N} . v_{z\,N}\right)$	thermal conductivity in arc and z-direction	macroscopic
82	$\rho_A := I_{N,A} \overset{N}{\star} \rho_N$	density in arc	macroscopic
83	$\hat{k}^{d,Fick}_{x\;A,S} := I_{N,A} \overset{N}{\star} \left(v_{x\,N} . \frac{\partial U_N}{\partial \mu_{N,S}} . (n_{N,S})^{-1}\right)$	Fick's diffusivity in arc and x-direction	macroscopic
84	$\hat{k}^{d,Fick}_{y\;A,S} := I_{N,A} \overset{N}{\star} \left(v_{y\,N} . \frac{\partial U_N}{\partial \mu_{N,S}} . (n_{N,S})^{-1}\right)$	Fick diffusivity in arc and y-direction	macroscopic
85	$\hat{k}^{d,Fick}_{z\;A,S} := I_{N,A} \overset{N}{\star} \left(v_{z\,N} . \frac{\partial U_N}{\partial \mu_{N,S}} . (n_{N,S})^{-1}\right)$	Fick diffusivity in arc and z-direction	macroscopic
86	$h_{A,S} := I_{N,A} \overset{N}{\star} h_{N,S}$	partial molar enthalpiies in arc	macroscopic
87	$\dot{n}^c_{x\,N,S} := F_{N,A} \overset{A}{\star} \hat{n}^c_{x\,A,S}$	accumulation of molar mass due to convection	macroscopic
88	$\dot{n}^d_{x\,N,S} := F_{N,A} \overset{A}{\star} \hat{n}^d_{x\,A,S}$	accumulation due to diffusion in x-direction	macroscopic
89	$\hat{n}^d_{x\,A,S} := k^d_{x\,A,S} . (A_{yz\,N} . F_{N,A}) \overset{N}{\star} \mu_{N,S}$	Fick diffusion flow in x-direction	macroscopic
90	$\hat{n}^d_{y\,A,S} := k^d_{y\,A,S} . (A_{yz\,N} . F_{N,A}) \overset{N}{\star} \mu_{N,S}$	Fick diffusion flow in y-direction	macroscopic
91	$\hat{n}^d_{z\,A,S} := k^d_{z\,A,S} . (A_{xy\,N} . F_{N,A}) \overset{N}{\star} \mu_{N,S}$	mass diffusion flow in z-direction	macroscopic

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no	equation	documentation	layer
92	$\dot{n}_{N,S} := \dot{n}^{c}_{x N,S} + \dot{n}^{d}_{x N,S} + V_{N} \, . \, \tilde{n}_{N,S}$	differential mass balance without reaction	macroscopic
93	$n_{N,S} := \int_{t^o}^{t^e} \dot{n}_{N,S} \; dt + n^{o}{}_{N,S}$	fundamental state – molar mass	macroscopic
94	$K_{N,K,p} := K^{o}{}_{K} \, . \, \mathbf{exp}\left(\left(-E^{a}{}_{K}\right) \, . \, \left(R \, . \, T_{N,p}\right)^{-1}\right)$	Arrhenius reaction "constant"	reactions
95	$\tilde{n}_{N,S,q} := A_{N,p,q} \overset{p}{\star} \left(N_{S,K} \overset{K}{\star} \left(K_{N,K,p} \, . \, \xi_{N,K,p}\right)\right)$	production from reaction set	reactions
98	$n^{o}{}_{N,S} := \mathbf{Instantiate}(n_{N,S}, \#)$	initial mass	macroscopic
99	$\dot{H}^{c}_{x N} := F_{N,A} \overset{A}{\star} \left(\hat{n}^{c}_{x A,S} \overset{S}{\star} h_{N,S}\right)$	enthalpy accumulation due to convective flow in x-direction	macroscopic
100	$\dot{H}^{d}_{x N} := F_{N,A} \overset{A}{\star} \left(\hat{n}^{d}_{x A,S} \overset{S}{\star} h_{N,S}\right)$	accumulation of enthalpy due to diffusional mass flow in x-direction	macroscopic
101	$\dot{H}^{d}_{y N} := F_{N,A} \overset{A}{\star} \left(\hat{n}^{d}_{y A,S} \overset{S}{\star} h_{N,S}\right)$	accumulation of enthalpy due to diffusional mass flow in y-direction	macroscopic
102	$\dot{H}^{d}_{z N} := F_{N,A} \overset{A}{\star} \left(\hat{n}^{d}_{z A,S} \overset{S}{\star} h_{N,S}\right)$	accumulation of enthalpy due to diffusional mass flow in z-direction	macroscopic
103	$\dot{q}_{x N} := F_{N,A} \overset{A}{\star} \hat{q}_{x A}$	accumulation due to heat flow in x-direction	macroscopic
104	$\dot{q}_{y N} := F_{N,A} \overset{A}{\star} \hat{q}_{y A}$	accumulation due to heat flow in y-direction	macroscopic
105	$\dot{q}_{z N} := F_{N,A} \overset{A}{\star} \hat{q}_{z A}$	accumulation due to heat flow in z-direction	macroscopic
106	$\hat{w}_{A} := \mathbf{Instantiate}(\hat{q}_{x A}, \#)$	a fixed work flow to start with	macroscopic
107	$\dot{n}^{d}_{y N,S} := F_{N,A} \overset{A}{\star} \hat{n}^{d}_{y A,S}$	accumulation due to diffusion in y-direction	macroscopic

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108	$\dot{n}^{d}_{z\,N,S} := F_{N,A} \overset{A}{\star} \hat{n}^{d}_{z\,A,S}$	accumulation due to diffusion in z-direction	macroscopic
109	$\dot{w}_{N} := F_{N,A} \overset{A}{\star} \hat{w}_{A}$	accumulation of enthalpy due to work flow	macroscopic
110	$\dot{H}_{N} := \dot{H}^{c}_{x\,N} + \dot{H}^{d}_{x\,N} + \dot{H}^{d}_{y\,N} + \dot{H}^{d}_{z\,N} + \dot{q}_{x\,N} + \dot{q}_{y\,N} + \dot{q}_{z\,N} + \dot{w}_{N}$	accumulation of enthalpy	macroscopic
111	${H^{o}}_{N} := \textbf{Instantiate}(H_{N}, \#)$	initial enthalpy	macroscopic
112	$H_{N} := \int_{t^{o}}^{t^{e}} \dot{H}_{N} \; dt + {H^{o}}_{N}$	Enthalpy	macroscopic
113	${U^{e}}_{N} := \left(C_{N}\right)^{-1} . U_{N}$	electrical potential – voltage	macroscopic
114	${I^{e}}_{N} := \frac{d\,C_{N}}{d\,t}$	current definition	macroscopic
115	${R^{e}}_{N} := \left({I^{e}}_{N}\right)^{-1} . {U^{e}}_{N}$	electrical resistant	macroscopic
116	${\dot{U}^{e}}_{A} := F_{N,A} \overset{A}{\star} \left({R^{e}}_{N} . {I^{e}}_{N}\right)$	Kirkhoffs first law	macroscopic
117	$C_{p\,N} := m_{N} . c_{p\,N}$	total heat capacity at constant pressure	macroscopic
119	${T^{ref}}_{N} := \textbf{Instantiate}(T_{N}, \#)$	reference temperature	macroscopic
120	$c_{p\,N} := \textbf{Instantiate}(c_{p\,N}, \#)$	constant specific heat capacity at constant pressure	macroscopic
121	$T_{N} := H_{N} . \left(C_{p\,N}\right)^{-1} + {T^{ref}}_{N}$	temperature from constant heat capacity	macroscopic
122	${T^{n}}_{N} := \textbf{Instantiate}(T_{N}, \#)$	value to norm measurement of temperature	macroscopic
123	$\bar{T}_{N} := T_{N} . \left({T^{n}}_{N}\right)^{-1}$	temperature measurement	macroscopic

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133	$\hat{m}_{N,A} := \hat{V}_A \,.\, \rho_N$	convective mass flow	macroscopic

12 Interface Link Equation

no	equation	documentation	layer
55	$\_c_{I,S} := {F^{source}}_{N,I} \overset{N}{\star} c_{N,S}$	interface equation	macroscopic –> reactions
56	$c_{N,S,p} := \left({F^{sink}}_{N,I} \cdot \_c_{I,S}\right) \overset{I}{\star} S_{I,p}$	interface equation	reactions
57	$\_x_{I,S} := {F^{source}}_{N,I} \overset{N}{\star} x_{N,S}$	interface equation	macroscopic –> reactions
58	$x_{N,S,p} := \left({F^{sink}}_{N,I} \cdot \_x_{I,S}\right) \overset{I}{\star} S_{I,p}$	interface equation	reactions
59	$\_T_{I} := {F^{source}}_{N,I} \overset{N}{\star} T_{N}$	interface equation	macroscopic –> reactions
60	$T_{N,p} := \left({F^{sink}}_{N,I} \cdot \_T_{I}\right) \overset{I}{\star} S_{I,p}$	interface equation	reactions
96	$\_np_{I,S} := \mathbf{reduceSum}\left(\left(\left({F^{source}}_{N,I} \overset{N}{\star} \tilde{n}_{N,S,q}\right) \cdot S_{I,q}\right), q\right)$	interface equation	reactions –> macroscopic
97	$\tilde{n}_{N,S} := {F^{source}}_{N,I} \overset{I}{\star} \_np_{I,S}$	interface equation	macroscopic
143	$\_mv\_I_{I} := {F^{source}}_{N,I} \overset{N}{\star} \bar{T}_{N}$	interface equation	macroscopic –> control
144	$T_meas_{N,p} := \left({F^{sink}}_{N,I} \cdot \_mv\_I_{I}\right) \overset{I}{\star} S_{I,p}$	interface equation	control